The Complete Treatise on Asymmetric Informational Ethics

A Formal Unification of Ethics, Information Theory, Decision Theory, Economics, Law, and Computer Science

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Abstract

We present *Asymmetric Informational Ethics* (AIE), a novel and formally complete school of ethical thought grounded in six foundational postulates. Ethics, on this account, is a dynamic system of *programs*—finite processes initiated by asymmetric information, driven by iterative investigation, and terminated by the joint exhaustion of informational asymmetries and outstanding decisions. We develop a rigorous formal apparatus comprising: a typed program grammar (AIEPN); first-order termination predicates; principal theorems including an AIE Non-Decidability result via Turing reduction; a process-tree spawning model; a three-tier conflict-resolution hierarchy; and an adjoined value theory synthesising preference revelation, *jus cogens*, and an ideal-endorsement condition. Algorithmic specifications, a state-machine formalisation, and PGF/Ti*k*Z diagrams are provided throughout. The framework is shown to subsume or relate to all major classical schools of ethics while generating novel, testable claims about the structure of moral life.

The treatise ends with "The End"

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1 Introduction

The history of systematic ethics is largely a history of *reduction*: the attempt to derive all moral obligations from a single supreme principle. Whether that principle is the maximisation of aggregate welfare (utilitarianism), the categorical imperative (Kantianism), the cultivation of virtue (Aristotelianism), or the terms of a rational social contract (contractarianism), each classical school seeks a monolithic foundation from which all particular obligations can be read off.

This treatise proposes a different starting point. Rather than beginning with a supreme value or principle, we begin with *epistemic structure*—with the observation that moral situations invariably involve parties who know different things. A patient and a physician, a witness and a court, an employee and the public: each pair is separated by informational asymmetries whose investigation and resolution constitutes the substance of the ethical encounter.

1.1 Motivation

Asymmetric Informational Ethics (AIE) is motivated by four observations.

(i) **Ethics is not static.** Moral situations have lifecycles: they arise, develop through inquiry, generate decisions, and—sometimes— resolve. Classical theories, which abstract away time and process, cannot capture this dynamism.

(ii) **Ethics is not monolithic.** A person simultaneously navigates obligations to family, employer, community, and posterity. These are not instances of a single master principle; they are structurally distinct programs running concurrently.

(iii) **Ethics is epistemically grounded.** Before a moral agent can act rightly, they must first *know* the morally relevant features of a situation. Classical theories treat facts as given and focus on what to do with them; AIE treats the acquisition and investigation of facts as itself the core ethical activity.

(iv) **Ethics is self-liquidating.** The purpose of ethical activity is to resolve itself—to reach a state in which the program is no longer needed. This feature, absent from all classical frameworks, has profound implications for the theory of moral progress.

1.2 Organisation

Section 2 establishes primitive concepts and formal definitions. Section 3 states and unpacks the six foundational postulates. Section 4 derives the principal theorems with full proofs. Section 5 resolves five open structural problems through contributions from philosophy, economics, computer science, and law. Section 6 introduces the AIEPN formal notation and type system. Section 7 provides algorithmic specifications. Section 8 develops the adjoined value theory. Section 9 situates AIE with respect to classical ethical schools. Section 10 discusses AI and machine-learning applications. Section 11 collects structural diagrams. The treatise closes with a glossary and bibliography.

2 Primitives, Definitions, and Notation

We proceed axiomatically. All complex notions are built from four undefined primitives.

2.1 Undefined Primitives

Agent An entity capable of holding propositional information and making selections among available actions.

Information A proposition or set of propositions about the world, together with its truth-value at a given time.

Decision A selection, by an agent, among a non-empty set of available actions at a time.

End A description of a terminal state toward which a process is directed.

2.2 Core Definitions

Definition 2.1 (Information State)**.** Let $I(a, t)$ denote the set of propositions held as true by agent a at time t. An *information state* is any such map $(a, t) \mapsto I(a, t)$.

Definition 2.2 (Asymmetric Information)**.** A condition of *asymmetric information* exists between agents a_1 and a_2 at time t if and only if $I(a_1, t) \neq I(a_2, t)$. An *asymmetry instance* is a triple $\alpha = (a_i, a_j, \varphi)$ where $\varphi \in I(a_i, t) \setminus I(a_j, t)$ for some t. The *total asymmetric information field* at time t is

$$A_\Omega(t) \;=\; \bigcup_{\substack{a_i,\, a_j \in \mathcal{A} \\ a_i \neq a_j}} \big(I(a_i, t) \setminus I(a_j, t)\big),$$

where $\mathcal{A}$ denotes the set of all agents.

Definition 2.3 (Ethics Program)**.** An *ethics program* P is a 4-tuple

$$P \;=\; \langle E,\; A,\; \Phi,\; D\rangle,$$

where E is the *end* (target terminal state); $A = \{\alpha_1, \alpha_2, \ldots\}$ is the *initiating asymmetry set*, a non-empty subset of A_Ω; Φ is the *investigation function*; and D is the *decision set* enabled by Φ.

Definition 2.4 (Investigation Function)**.** The *investigation function* Φ of program P is an iterated map

$$A^{(0)} \xrightarrow{\Phi} A^{(1)} \xrightarrow{\Phi} A^{(2)} \xrightarrow{\Phi} \cdots \xrightarrow{\Phi} A^{(n)}.$$

Each application may reduce, expand, or resolve elements of the asymmetry set; convergence to $\emptyset$ is not assumed at any finite step.

Definition 2.5 (Decision Enablement)**.** A decision $d \in D$ is *enabled* by program P at time t, written $d \in \delta(P, t)$, if and only if some iteration $\Phi^{(k)}$, $k \leq t$, has produced sufficient reduction in $A^{(0)}$ to render d actionable:

$$d \in \delta(P, t) \iff \exists\, k \leq t : \left|A^{(k)}\right| < \left|A^{(0)}\right| \;\text{ and }\; d \text{ is actionable under } A^{(k)}.$$

Definition 2.6 (Program Termination)**.** Program P *terminates* at time t^* if and only if

$$\mathsf{TERM}(P) \iff \underbrace{A^{(t^*)} = \emptyset}_{\text{(i) no residual asymmetry}} \;\wedge\; \underbrace{\delta(P, t^*) = \emptyset}_{\text{(ii) no outstanding decisions}}. \tag{1}$$

Both conjuncts are necessary; neither alone is sufficient.

Definition 2.7 (Ethics System)**.** The *ethics system* is the (possibly infinite) collection $\mathbf{\Sigma} = \{P_1, P_2, P_3, \ldots\}$. Programs in $\mathbf{\Sigma}$ may be concurrent, sequential, nested, or independent.

Definition 2.8 (Global Termination)**.** The ethics system $\mathbf{\Sigma}$ terminates if and only if every member program terminates:

$$\mathsf{TERM}(\mathbf{\Sigma}) \iff \forall\, P_i \in \mathbf{\Sigma} : \mathsf{TERM}(P_i).$$

2.3 Program Weight

Conflict resolution (Section 5.2) requires a cardinal measure of program importance.

Definition 2.9 (Weight Function)**.** The *weight* of program P_i is

$$\omega(P_i) \;=\; s_i \;\cdot\; u_i \;\cdot\; (1 + r_i), \tag{2}$$

where $s_i \in \mathbb{N}$ is the *scope* (number of agents affected); $u_i \in [0, 1]$ is the *urgency* (rate of harm accrual if unresolved); and $r_i \in [0, 1]$ is the *reversibility penalty* (larger when unresolved harm is irreversible).

3 The Six Foundational Postulates

Postulate 1 (Compositional Structure)**.** Ethics consists of a structured collection of programs, each with a distinct end:

$$\mathbf{\Sigma} = \{P_i\}, \qquad P_i = \langle E_i,\, A_i,\, \Phi_i,\, D_i\rangle, \qquad E_i \text{ distinct for each } i.$$

Postulate 2 (Asymmetric Initiation)**.** An ethics program P_i is initiated at time t if and only if the total asymmetric information field is non-empty and the program's initiating asymmetry is drawn from it:

$$P_i \text{ initiated at } t \iff A_\Omega(t) \neq \emptyset \;\wedge\; A_i \subseteq A_\Omega(t).$$

Postulate 3 (Investigative Dynamics)**.** Every ethics program proceeds through iterative application of its investigation function to its asymmetry set:

$$A_i^{(0)} \xrightarrow{\Phi_i} A_i^{(1)} \xrightarrow{\Phi_i} \cdots \xrightarrow{\Phi_i} A_i^{(n)}.$$

Postulate 4 (Decision Enablement)**.** Every decision in a program's decision set is enabled at some point during the investigative process: $\forall\, d \in D_i,\ \exists\, t : d \in \delta(P_i, t)$.

Postulate 5 (Conjunctive Termination)**.** Program P_i terminates at t^* if and only if $A_i(t^*) = \emptyset$ and $\delta(P_i, t^*) = \emptyset$ (cf. equation (1)).

Postulate 6 (Global Termination)**.** Ethics as a whole terminates if and only if every program in $\mathbf{\Sigma}$ terminates (cf. Definition 2.8).

Figure 1 illustrates the lifecycle of a single ethics program under the six postulates.

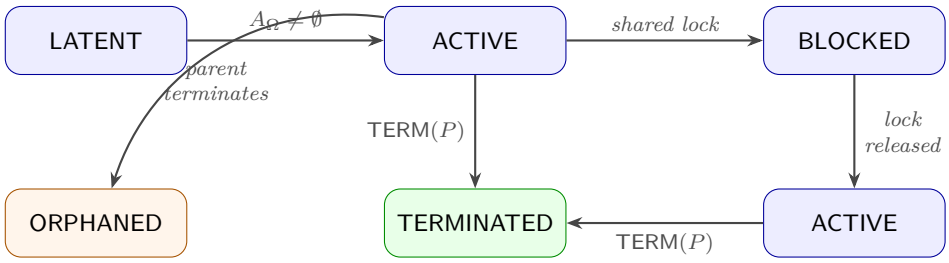

Figure 1: Lifecycle state machine of an AIE ethics program. Transitions are labelled with their enabling conditions. The ORPHANED state arises when a weakly-nested child program's parent terminates before the child does.

4 Principal Theorems

Theorem 4.1 (Ethics Requires Ignorance)**.** *If $A_\Omega(t) = \emptyset$ for all t, then $\mathbf{\Sigma} = \emptyset$.*

Proof. By Postulate 2, every program P_i requires $A_i \subseteq A_\Omega(t)$ for some t. If $A_\Omega(t) = \emptyset$ for all t, no A_i can be non-empty, so no program can be initiated. Hence $\mathbf{\Sigma} = \emptyset$. □

Corollary 4.2. *Perfect, universally shared information precludes ethics entirely. Moral life is constitutively dependent on epistemic inequality.*

Theorem 4.3 (Decisions Cannot Precede Investigation)**.** *For any program $P = \langle E, A, \Phi, D\rangle$, no decision in D is enabled at $t = 0$.*

Proof. By Definition 2.5, $d \in \delta(P, t)$ requires some $k \leq t$ such that $|A^{(k)}| < |A^{(0)}|$. At $t = 0$ no application of Φ has occurred, so the strict inequality $|A^{(0)}| < |A^{(0)}|$ cannot hold. Hence $\delta(P, 0) = \emptyset$. □

Corollary 4.4. *Ethical impulsiveness—deciding without investigation—is a formal violation of program structure. It is not merely unwise; it is pre-ethical.*

Theorem 4.5 (Partial Resolution Does Not Terminate a Program)**.** *For any program P and time t:*

$$A^{(t)} = \emptyset \;\not\Rightarrow\; \mathsf{TERM}(P) \qquad \textit{and} \qquad \delta(P, t) = \emptyset \;\not\Rightarrow\; \mathsf{TERM}(P).$$

Proof. Termination requires both conjuncts of (1). A program may satisfy $A^{(t)} = \emptyset$ while $\delta(P, t) \neq \emptyset$ (asymmetries resolved but decisions still pending), and vice versa. Each conjunct is necessary but not sufficient. □

Theorem 4.6 (Investigation Can Expand Before Contracting)**.** *There exists a program P and index k such that $|A^{(k+1)}| > |A^{(k)}|$.*

Proof. By Definition 2.4, Φ maps $A^{(k)}$ to a *transformed* set, not necessarily a smaller one. A step may produce an $\mathsf{EXPANDS}(\alpha')$ operation, introducing a new asymmetry $\alpha' \notin A^{(k)}$ into $A^{(k+1)}$. Writing R for resolved and N for newly revealed asymmetries:

$$|A^{(k+1)}| \;=\; |A^{(k)}| - |R| + |N|.$$

Whenever $|N| > |R|$ the set grows. Such expansion arises naturally when investigation uncovers hidden informational structure (e.g. a regulatory gap revealed during contract adjudication). □

Corollary 4.7. *Ethical investigation can make a moral situation more complex before resolving it. A growing program is not malfunctioning—it is working correctly.*

Theorem 4.8 (Ethics is Potentially Non-Terminating)**.** *In a world of sufficient agent complexity,* $\mathbf{\Sigma}$ *need never terminate.*

Proof. Let $\lambda(t)$ and $\mu(t)$ denote the instantaneous rates of program initiation and termination respectively. By Postulate 2, new programs are initiated whenever new asymmetric information arises. If $\lambda(t) \geq \mu(t)$ for all $t > 0$, then $|\mathbf{\Sigma}(t)|$ is non-decreasing and $\mathbf{\Sigma}$ does not terminate. In any world where agents continuously interact and generate information—as in any realistic social environment—the condition $\lambda \geq \mu$ holds generically. □

Theorem 4.9 (AIE Non-Decidability)**.** *There is no general procedure that can determine, from the initial conditions* $P = \langle E, A^{(0)}, \Phi, D\rangle$ *alone, whether* P *will terminate.*

Proof. We reduce to Turing's Halting Problem [20]. Encode P as a Turing machine M_P whose tape alphabet represents asymmetry status, whose transition function encodes Φ, and whose halting state corresponds to $\mathsf{TERM}(P)$. This encoding is computable. By the Halting Problem [20], no algorithm decides whether an arbitrary Turing machine halts on all inputs. If $\mathsf{TERM}(P)$ were decidable in general, the Halting Problem would be decidable—a contradiction. Hence $\mathsf{TERM}(P)$ is undecidable in general. □

Corollary 4.10. *No ethical situation can be guaranteed to resolve from its starting conditions alone. Resolution depends irreducibly on the conduct of agents throughout the program's lifetime. Ethics cannot be pre-computed.*

5 Resolving the Open Structural Problems

Five structural problems arise naturally from the formal apparatus. Each is resolved through a targeted disciplinary contribution.

5.1 The Scoping Problem

What formal criteria determine which asymmetries belong to which program, and can programs share asymmetric information?

Relevance Criterion (Philosophy + Computer Science)

Definition 5.1 (Functional Relevance)**.** Asymmetry instance α is *functionally relevant* to program P if and only if its inclusion would alter either the investigation function's output or the set of enabled decisions:

$$\Phi\big(A \cup \{\alpha\}\big) \neq \Phi(A) \quad \vee \quad \delta\big(P,\, A \cup \{\alpha\}\big) \neq \delta(P,\, A).$$

Definition 5.2 (Shared Asymmetry Set)**.** For programs P_i and P_j, the *shared asymmetry set* is $A_s = A_i \cap A_j \neq \emptyset$. Elements of A_s receive status SHARED_LOCK and must be investigated as *critical sections*, coordinated to prevent race conditions analogous to concurrent shared-memory access [19].

Algorithm 1 formalises the scoping procedure.

Algorithm 1 Asymmetry Scoping Procedure

Require: New asymmetry α; active program set $\mathbf{\Sigma}$; coordination register $\mathcal{R}$
Ensure: α assigned to all functionally relevant programs; shared lock registered if $|\mathsf{assigned}| > 1$

```
assigned ← ∅
for each P_k ∈ Σ do
    if Φ_k(A_k ∪ {α}) ≠ Φ_k(A_k)  or  δ(P_k, A_k ∪ {α}) ≠ δ(P_k, A_k) then
        A_k ← A_k ∪ {α}
        assigned ← assigned ∪ {P_k}
    end if
end for
if |assigned| > 1 then
    α.status ← SHARED_LOCK
    R.register(α, assigned)
end if
return assigned
```

5.2 The Conflict Problem

When programs P_i and P_j have incompatible ends E_i and E_j, what meta-principle governs priority?

Three-Tier Resolution Hierarchy (Economics + Law + Philosophy)

Tier 1: Pareto Efficiency. Resolution R is *Pareto-preferred* if no alternative R' improves one program's prospects without worsening the other's:

$$R \text{ is Pareto-preferred} \iff \nexists R' : V(E_j \mid R') > V(E_j \mid R) \wedge V(E_i \mid R') \geq V(E_i \mid R),$$

where $V(E \mid R)$ measures the value-to-program of resolution R. Many apparent conflicts dissolve at Tier 1, since they are artefacts of asymmetric information rather than genuine value incompatibility.

Tier 2: Weighted Priority. For genuine conflicts surviving Tier 1, the program with higher weight takes lexical priority: $P_i \succ P_j \iff \omega(P_i) > \omega(P_j)$. This is structurally analogous to Rawlsian lexical ordering [14] but grounded in informational and decisional properties rather than abstract principles of justice.

Tier 3: Adjudication. When Tiers 1 and 2 fail to yield a determinate outcome, AIE invokes *adjudication* by a neutral third party, drawing on the legal tradition of adversarial presentation, rules of evidence, precedent, and proportionality review [8].

Definition 5.3 (Three-Tier Meta-Principle)**.** For programs P_i and P_j with incompatible ends, apply Tier 1 (Pareto), then Tier 2 (weighted priority), then Tier 3 (adjudication), in strict sequence. A resolution is reached at the first tier yielding a determinate outcome.

Table 1: Three-tier conflict resolution hierarchy.

Tier	Name	Discipline	Resolution criterion
1	Pareto Efficiency	Economics	No program worsened by the resolution
2	Weighted Priority	Econ./Philosophy	$\omega(P_i) > \omega(P_j)$ per (2)
3	Adjudication	Law	Neutral party; precedent; proportionality

5.3 The Convergence Problem

Under what conditions is convergence of Φ to $A^{(n)} = \emptyset$ formally guaranteed?

Fixed-Point Analysis (Computer Science + Philosophy)

Reframe Φ as operating on the complete lattice $(\mathcal{P}(A_\Omega), \subseteq)$. By Tarski's fixed-point theorem [18], any monotone function on a complete lattice has a fixed point. Convergence to $\emptyset$ is the least fixed point of Φ.

Theorem 5.4 (Convergence Conditions)**.** *Φ converges to $A^{(n)} = \emptyset$ if and only if:*

(C1) ***Finite Asymmetry.*** *$|A_\Omega|$ is finite, or bounded within the program's domain.*

(C2) ***Investigative Honesty.*** *Φ is applied without strategic introduction of new asymmetries by any agent.*

(C3) ***Decisional Closure.*** *Every enabled decision is eventually made rather than deferred indefinitely.*

Proof. (Necessity.) If (C1) fails, $|A_\Omega|$ may be unbounded and Φ may never exhaust all asymmetries. If (C2) fails, agents may engineer $|N| > |R|$ at each step (Theorem 4.6), preventing convergence. If (C3) fails, $\delta(P, t) \neq \emptyset$ persists indefinitely, blocking termination by Postulate 5. *(Sufficiency.)* Under (C1)–(C3), $|A^{(k)}|$ is non-increasing after a finite expansion phase bounded by (C1); each honest step reduces at least one asymmetry without replacement; and decisions are made as enabled. Hence Φ reaches $A^{(n)} = \emptyset$ in finitely many steps. □

5.4 The Spawning Problem

Can investigation within P_i initiate a new program P_j, and if so, is P_j nested within or independent of P_i?

Process Trees and Ancillary Jurisdiction (CS + Law)

Definition 5.5 (Spawning)**.** Program P_j is *spawned* by P_i at iteration k if $A_j \subseteq \Phi_i^{(k)}$—that is, P_j's initiating asymmetry was produced by investigation within P_i.

Definition 5.6 (Nesting Kind)**.** A spawned program P_j is *strongly nested* in P_i if resolving E_j is necessary and material to resolving E_i; P_i's termination is blocked until P_j terminates. Otherwise P_j is *weakly nested* and runs independently.

The strong/weak distinction mirrors the legal doctrine of *ancillary jurisdiction* [8]: a court may hear subsidiary matters only when they are necessary and material to the primary claim.

Definition 5.7 (Orphan Program)**.** If parent P_i terminates while weakly-nested child P_j has not, P_j becomes an *orphan program*, persisting in $\mathbf{\Sigma}$ by virtue of its own matter—analogous to *in rem* jurisdiction.

Figure 2 illustrates a three-level process tree.

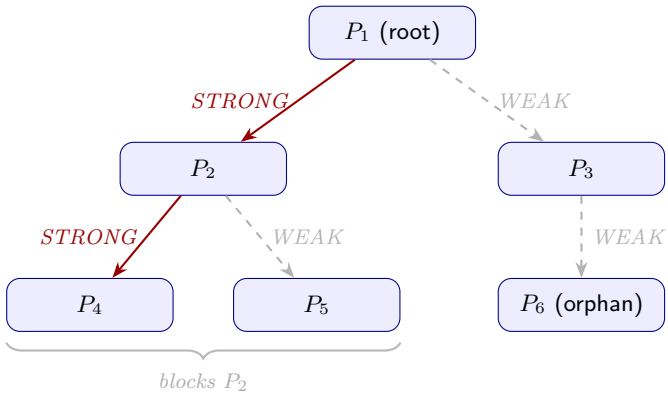

Figure 2: Program process tree. Solid arrows indicate strong nesting (blocked parent); dashed arrows indicate weak nesting (independent parent). P_6 is orphaned when P_3 terminates.

5.5 The Valuation Problem

AIE is structurally complete but normatively neutral: it describes the form of ethics without specifying what makes an end E worth pursuing. An adjoined value theory is required.

The adjoined value theory is developed in full in Section 8.

6 The Formal Notation: AIEPN

We introduce the *AIE Program Notation* (AIEPN), a formal language for representing ethics programs as computable structures. AIEPN programs admit parsing, type-checking against four invariants, and simulation as finite-state machines.

6.1 Grammar

The grammar is presented in extended Backus–Naur form. Terminal keywords are in **bold**; non-terminals in ⟨*angle brackets*⟩.

⟨*System*⟩	::=	**SIGMA** { ⟨*Prog*⟩$^+$ }
⟨*Prog*⟩	::=	**PROG** ⟨*Id*⟩ **END**:E **ASYM**:A **PHI**:Φ **DEC**:D **STATUS**:S [**WEIGHT**:W] [**PARENT**:p] [**CHILDREN**:N] **END_PROG**
⟨*AEntry*⟩	::=	⟨*Id*⟩ : ⟨*Pair*⟩ × ⟨*Prop*⟩ [**WT**:f] [**SHARED**:⟨*Id*⟩$^+$] **STATUS**:S
⟨*Status*⟩	::=	**OPEN** | **UNDER_INV** | **RESOLVED** | **SHARED_LOCK**
⟨*State*⟩	::=	**LATENT** | **ACTIVE** | **BLOCKED** | **TERMINATED** | **ORPHANED**
⟨*Step*⟩	::=	**STEP** ⟨*Id*⟩ **TO**:⟨*Id*⟩$^+$ **METHOD**:M **PRODUCES**:Δ [**SPAWNS**:σ]
M	::=	**INQUIRY** | **DELIBERATION** | **ADJUDICATION** | **NEGOTIATION** | **EVIDENCE**
Δ	::=	**REDUCES**(⟨*Id*⟩$^+$) | **EXPANDS**(⟨*AEntry*⟩$^+$) | **RESOLVES**(⟨*Id*⟩$^+$)
W	::=	{ **SCOPE**:n, **URGENCY**:u, **REVS**:r }, $n \in \mathbb{N}$, $u, r \in [0, 1]$

Non-terminals: Φ = InvestFn; A = AsymSet; D = DecSet; N = NestSpec; p = parent Id.
Abbreviations used above: **EVIDENCE** = **EVIDENCE_GATHERING**; **REVS** = **REVERSIBILITY**.

6.2 Type System and Invariants

Table 2: AIEPN type system.

Type	Definition	Role
`Agent`	`String`	Named entity with information state
`AgentPair`	`Agent` $\times$ `Agent`	Ordered pair: (information holder, lacking)
`Proposition`	`String`	Natural-language or formal statement
`Float[0,1]`	$\mathbb{R} \cap [0,1]$	Bounded real for salience weights
`AsymEntry`	Record (6 fields)	Single asymmetry instance with status
`PhiStep`	Record (5 fields)	One iteration step of investigation
`DecEntry`	Record (3 fields)	Decision with enablement and status
`Program`	Record (9 fields)	Complete ethics program and metadata

The four enforced type invariants are:

(I1) *Decision integrity.* Every decision references a valid Φ step:

$$\forall\, d \in P.dec,\ s \in d.enabled_by \ \Rightarrow\ s \in \{s'.id \mid s' \in P.phi\}.$$

(I2) *Shared-asymmetry consistency.* Every shared asymmetry is mutually registered in both programs:

$$\forall\, \alpha \in P_i.asym,\ P_j \in \alpha.shared_with \ \Rightarrow\ \alpha.id \in \{\beta.id \mid \beta \in P_j.asym\}.$$

(I3) *Strong-child blocking.* For all c with $P.children[c] = STRONG$:

$$\neg\mathsf{TERM}(P) \ \vee\ \mathsf{TERM}(\mathbf{\Sigma}[c]).$$

(I4) *Computed weight.* $\omega(P) := P.scope \cdot P.urgency \cdot (1 + P.reversibility)$.

7 Algorithmic Specifications

7.1 Core Inference Engine

Algorithm 2 specifies the AIE inference engine—the central computational process governing a running ethics system.

Algorithm 2 AIE Inference Engine

Require: Ethics system $\mathbf{\Sigma}$; agent set $\mathcal{A}$; horizon T
Ensure: Updated $\mathbf{\Sigma}$ at termination or horizon

$t \leftarrow 0$
while $t < T$ **and** $\neg\,\mathsf{TERM}(\mathbf{\Sigma})$ **do**
  $A_\Omega(t) \leftarrow \bigcup_{a_i \neq a_j} \big(I(a_i,t) \setminus I(a_j,t)\big)$ ▷ recompute total asymmetry field
  for each $\alpha \in A_\Omega(t)$ **do**
    $\mathsf{Scope}(\alpha,\ \mathbf{\Sigma})$ ▷ Algorithm 1
  end for
  for each $P_k \in \mathbf{\Sigma}$ with $P_k.status = \textbf{ACTIVE}$ **do**
    $A_k^{(t+1)} \leftarrow \Phi_k\Big(A_k^{(t)}\Big)$
    $\delta(P_k,\ t{+}1) \leftarrow \mathsf{EnabledDecisions}(P_k,\ A_k^{(t+1)})$
    if P_k has a spawn operation at this step **then**
      $\mathbf{\Sigma} \leftarrow \mathbf{\Sigma} \cup \{\mathsf{Spawn}(P_k)\}$
    end if
    if $\mathsf{TERM}(P_k)$ **then**
      $P_k.status \leftarrow \textbf{TERMINATED}$
      $\mathsf{OrphanCheck}(\mathbf{\Sigma},\ P_k)$
    end if
  end for
  $\mathsf{ResolveConflicts}(\mathbf{\Sigma})$ ▷ Section 5.2
  $t \leftarrow t + 1$
end while
return $\mathbf{\Sigma}$

7.2 Deadlock Detection

Algorithm 3 Deadlock Detection and Escalation

Require: Program P; stagnation threshold $\theta \in \mathbb{N}$
Ensure: Escalation to Tier 3 adjudication if deadlocked

```
k_stall ← 0;   A_prev ← A^(0)
for each iteration k = 1, 2, ... do
    A^(k) ← Φ(A^(k−1))
    if A^(k) = A_prev then
        k_stall ← k_stall + 1
    else
        k_stall ← 0;   A_prev ← A^(k)
    end if
    if k_stall ≥ θ then
        flag P as deadlocked
        escalate to Tier 3 adjudication                ▷ Section 5.2
        break
    end if
end for
```

8 The Adjoined Value Theory

The formal apparatus of AIE is structurally complete but normatively neutral: it describes the *form* of ethics without specifying what makes an end E worth pursuing. An adjoined value theory supplies this normative content.

8.1 Three-Component Value Theory

Component 1: Endogenous Value Emergence (Economics)

Drawing on Samuelson's revealed preference theory [15] and Varian's formalisation [21], values are discovered endogenously through observable patterns of program pursuit rather than stipulated externally.

Definition 8.1 (Revealed Value Function)**.** The *revealed value function* of agent a is

$$V(a) \;=\; \big\{E_i : a \text{ has consistently initiated and sustained programs toward } E_i\big\}.$$

An agent who claims to hold value E but never initiates programs toward E is revealed, formally, not to hold it. The definition is empirically falsifiable, grounding ethics in observable behaviour.

Component 2: Inviolable Constraints (Law)

Endogenous emergence alone cannot exclude monstrous but consistently pursued ends. We adjoin a finite set of lexically prior constraints.

Definition 8.2 (Constraint Set)**.** Let $\mathcal{C} = \{c_1, \ldots, c_m\}$ be a set of *peremptory constraints* (analogous to *jus cogens* norms [8]) such that no end E_i is admissible if its pursuit requires violating any $c_j \in \mathcal{C}$. The set $\mathcal{C}$ must be: (a) *finite*; (b) *lexically prior*, never overridden by $\omega(P)$ or Pareto arguments; and (c) *publicly known*, enabling a priori program evaluation.

Component 3: Ideal Endorsement (Philosophy)

Drawing on Habermas's discourse ethics [7] and Scanlon's contractualism [16], an end is value-valid only if it would survive full informational disclosure to all affected parties.

Definition 8.3 (Ideal Endorsement Condition)**.** End E_i is *value-valid* if and only if every affected agent would endorse E_i given full information:

$$\forall\, a_k \text{ affected by } P_i :\; a_k \text{ would endorse } E_i \text{ given } I^*(a_k, t^*),$$

where $I^*(a_k, t^*)$ is the information state at program termination— the state of maximal informational symmetry.

Ends that depend on *maintaining* asymmetric information for their pursuance are formally *value-suspect*; ends that survive full informational disclosure are *value-robust*.

8.2 The Legitimacy Theorem

Theorem 8.4 (Legitimacy of an End)**.** *An end E_i is a* legitimate target *for an ethics program if and only if all three conditions hold:*

(V1) *No constraint $c_j \in \mathcal{C}$ is violated by pursuing E_i;*

(V2) *$E_i \in V(a)$ for the initiating agent a;*

(V3) *E_i satisfies the ideal endorsement condition.*

All three are necessary; none alone is sufficient.

Figure 3 represents the three-component value theory.

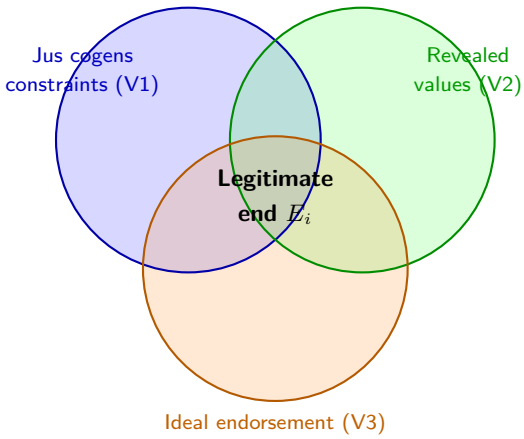

Figure 3: The three-component value theory. A legitimate end must lie at the intersection of all three regions: not violating any peremptory constraint (V1), consistent with the agent's revealed values (V2), and endorsable by all affected parties under conditions of full informational symmetry (V3).

9 AIE and Classical Ethical Schools

Table 3 situates AIE with respect to the seven major classical schools. AIE does not refute these traditions; rather, it provides a more general framework within which each appears as a special case or partial instantiation.

Table 3: AIE and classical ethical schools.

School	**Core claim**	**AIE perspective**
Consequentialism [11]	Right action maximises aggregate welfare	E = welfare maximisation; AIE requires investigation before assessment
Deontology [9]	Act on universalisable maxims; respect persons as ends	Pre-scoped programs with fixed E, D; AIE challenges their pre-investigative character
Virtue Ethics [1]	Cultivate stable dispositions of character	Investigative rigour and honest scoping are virtues in AIE; vice is a failure of investigation
Contractarianism [14]	Morality is principles rational agents would agree to	Informational asymmetry between parties is what contracts resolve; AIE generalises this
Care Ethics [6]	Morality arises from particular caring relationships	Relational asymmetries of need and vulnerability are a primary source of AIE programs
Natural Law	Moral norms are grounded in human nature and reason	The constraint set $\mathcal{C}$ may be populated via natural law theory without altering AIE's structure
Pragmatism [4]	Moral beliefs are hypotheses tested by experience	AIE's iterative Φ formalises pragmatist moral inquiry; deadlock triggers revision

10 AI and Machine-Learning Applications

AIE has direct and non-trivial applications to machine ethics, AI alignment, and the design of autonomous moral agents.

10.1 Moral Agent as Inference Engine

An AIE moral agent is a classifier–planner hybrid. The classification component detects asymmetric information warranting a new program; the planning component applies Φ iteratively and enables decisions. Figure 4 illustrates the architecture.

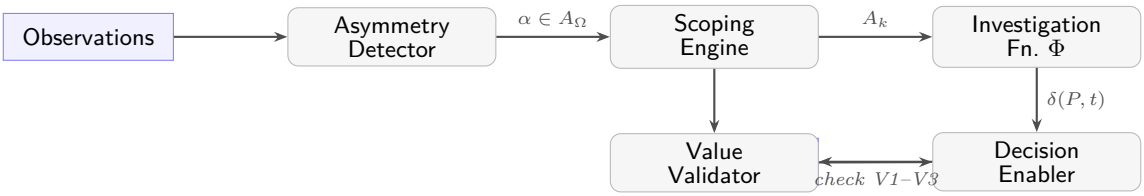

Figure 4: Architecture of an AIE moral agent as an inference engine. Observations feed the asymmetry detector; scoped asymmetries drive the investigation function Φ; enabled decisions pass through the value validator (conditions V1–V3, Theorem 8.4) before action.

10.2 Ethical Regret as a Learning Objective

Define the *ethical regret* of a moral agent up to horizon T as

$$\mathcal{L}(T) \;=\; \sum_{P_i \in \mathbf{\Sigma}} \int_0^T \left|A_i^{(t)}\right| \cdot u_i \, \mathrm{d}t, \tag{3}$$

where u_i is the urgency weight of program P_i. Minimising $\mathcal{L}(T)$ is a well-posed optimisation objective: it penalises accumulated uninvestigated asymmetry, weighted by moral urgency. The functional is non-negative, differentiable in u_i, and equals zero if and only if $\mathbf{\Sigma}$ terminates before T.

10.3 Convergence Profiles

Figure 5 plots $|A^{(k)}|$ under three representative convergence profiles, illustrating monotone reduction, expansion-then-reduction (Theorem 4.6), and deadlock.

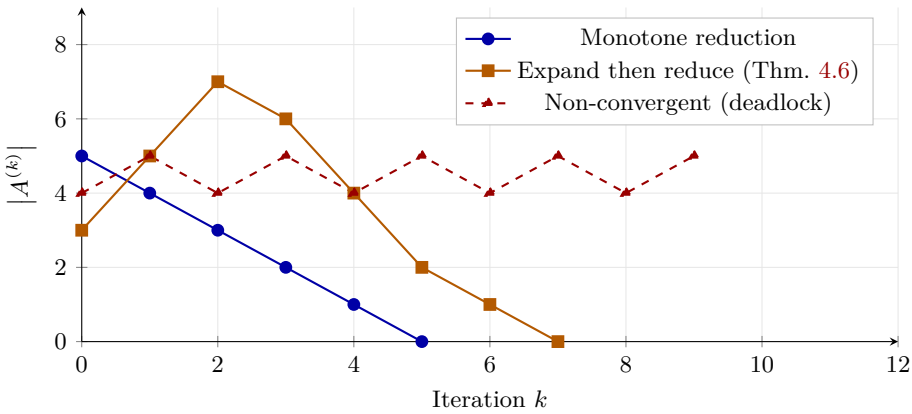

Figure 5: Asymmetry set size $|A^{(k)}|$ under three convergence profiles. The orange curve illustrates Theorem 4.6: investigation expands the asymmetry set before converging. The dashed red curve triggers Algorithm 3 after θ stagnant iterations.

11 Complete Structural Diagrams

11.1 Layered System Architecture

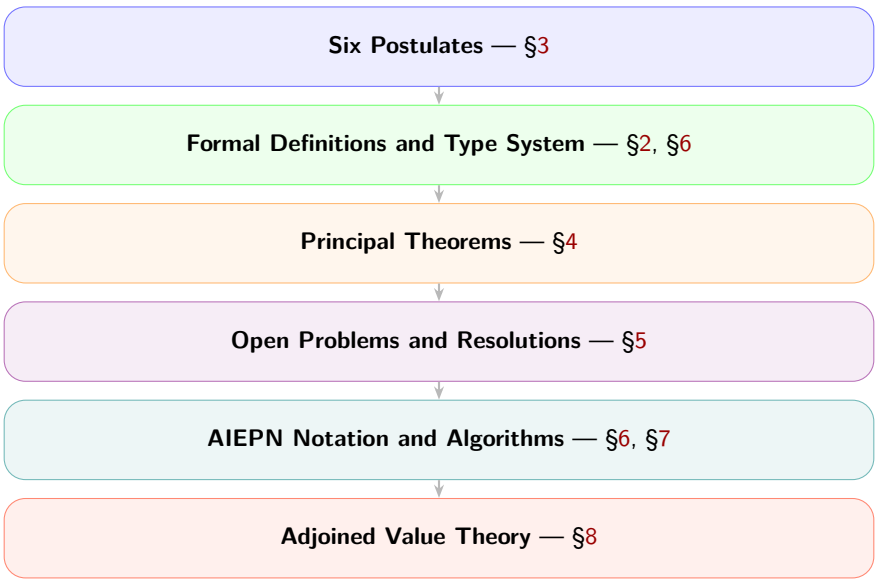

Figure 6: Layered architecture of the AIE formal system. Each layer depends logically on all layers above it; the system is stratified from pure postulates down to computable notation and normative value theory.

11.2 Φ-Iteration and Decision Enablement

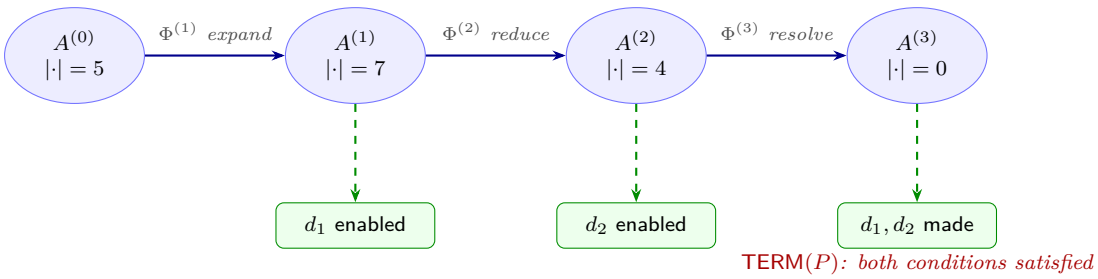

Figure 7: Φ-iteration and decision enablement for an illustrative program. The asymmetry set first expands at iteration 1 (Theorem 4.6), then contracts to zero. Decisions d_1 and d_2 are enabled incrementally; termination is declared only when both $A^{(3)} = \emptyset$ and all decisions are made.

12 Discussion and Conclusions

12.1 Summary of Contributions

We have presented Asymmetric Informational Ethics as a formally complete and multi-disciplinarily grounded school of ethical thought. Its principal contributions are:

(1) A *process-theoretic ontology*: moral situations are programs with lifecycles, not static evaluations of actions or characters.

(2) A *formal grammar* (AIEPN) and *type system* with four enforced invariants, enabling ethics programs to be written, type-checked, and simulated as finite-state machines.

(3) The *AIE Non-Decidability Theorem* (Theorem 4.9), establishing via Turing reduction that no ethical situation can be guaranteed to resolve from its initial conditions alone.

(4) A *multi-disciplinary resolution* of five structural open problems, drawing on philosophy, economics, computer science, and law.

(5) The *Legitimacy Theorem* (Theorem 8.4), with three necessary and jointly sufficient conditions for an end's legitimacy, and the *ethical regret functional* (3), a well-posed ML training objective for moral agents.

12.2 Limitations and Future Work

Several important problems remain open.

(a) *Population of $\mathcal{C}$.* The constraint set is characterised axiomatically but not populated. Specifying which peremptory norms belong to $\mathcal{C}$, and how they interact, remains an open normative question.

(b) *Quantitative investigation methods.* AIEPN enumerates investigation methods but does not formalise their relative effectiveness or computational cost.

(c) *Multi-agent coordination.* When multiple agents apply Φ concurrently to shared asymmetries, the shared-lock mechanism prevents race conditions but does not specify optimal coordination strategies.

(d) *Empirical validation.* The ethical regret functional $\mathcal{L}(T)$ is a testable objective; its behaviour in practical AI alignment systems deserves empirical study.

12.3 Conclusion

AIE offers a genuinely novel and formally rigorous approach to ethics—one that begins not with a supreme value or principle but with the epistemic structure of moral situations. Its central insight is at once simple and generative: *ethics arises from ignorance.* Where all parties share all relevant knowledge equally, there is nothing to investigate, no decision to enable, and no program to run. Moral life is the systematic effort to reduce the asymmetries that separate us—and the self-liquidating character of that effort is, perhaps, its most distinctively hopeful feature.

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Glossary

Agent
An entity capable of holding information and making decisions. Undefined primitive of AIE.

AIE
Asymmetric Informational Ethics; the school of ethical thought developed in this treatise.

AIEPN
AIE Program Notation; the formal grammar introduced in §6 for representing ethics programs as computable structures.

Asymmetric Information
A state in which two or more agents hold different information states: $I(a_1, t) \neq I(a_2, t)$.

Convergence
The process by which Φ reduces $A^{(k)}$ to $\emptyset$; guaranteed under conditions (C1)–(C3) of Theorem 5.4.

Deadlock
A state in which Φ produces no change in A over θ consecutive iterations; triggers Tier 3 adjudication via Algorithm 3.

Decision
A selection among available actions; enabled by investigation and unable to precede it (Theorem 4.3).

End (E)
The target terminal state of an ethics program; every program has exactly one (Postulate 1).

Ethical Regret
The functional $\mathcal{L}(T)$ measuring accumulated uninvestigated asymmetry weighted by urgency (equation (3)).

Ethics Program
The 4-tuple $P = \langle E, A, \Phi, D \rangle$; the fundamental unit of AIE (Definition 2.3).

Ethics System ($\boldsymbol{\Sigma}$)
The set of all active ethics programs.

Ideal Endorsement
The requirement that end E_i would be endorsed by all affected agents under conditions of full informational symmetry.

Investigation Fn. (Φ)
The iterated map transforming the asymmetry set; the engine of ethical inquiry (Definition 2.4).

Jus Cogens
Peremptory norms from which no derogation is permitted; here adapted as the content of the constraint set $\mathcal{C}$.

Moral Agent
An agent competent in AIE: one who detects asymmetric information, scopes programs correctly, applies Φ honestly, times decisions appropriately, and monitors termination conditions.

Nesting
Strong: child termination required before parent. Weak: child runs independently of parent.

Orphan Program
A weakly-nested child whose parent has terminated; persists in $\boldsymbol{\Sigma}$ by virtue of its own matter.

Pareto Efficiency
Tier-1 conflict resolution: prefer resolutions that worsen no program's prospects.

Process Tree
The directed acyclic graph of parent–child spawning relationships among ethics programs.

Revealed Value Fn. (V)
The set of ends consistently pursued by agent a across programs; basis of Component 1 of the value theory.

Scoping
Assignment of asymmetry instances to programs based on functional relevance (Algorithm 1).

Shared Lock
Status of an asymmetry belonging to multiple programs; requires coordinated investigation.

Spawning
Initiation of a new program P_j whose asymmetry was produced by investigation within P_i.

Termination
The state in which $A^{(t^*)} = \emptyset$ and $\delta(P, t^*) = \emptyset$ simultaneously (Definition 2.6).

Total Asymmetric Field (A_Ω)
The union of all pairwise informational differences across all agents at a given time.

Weight Function (ω)
Cardinal measure of program importance: $\omega(P) = s \cdot u \cdot (1 + r)$ (Definition 2.9).

The End